Vijaya Krishna Yalavarthi

www.ismll.uni-hildesheim.de/personen/yalavarthi
yalavarthi@ismll.de yalavarthivk
vijaya-krishna-yalavarthi-27a4414b Google Scholar
Information Systems and Machine Learning Lab, University of Hildesheim, Germany

Research Interests

Deep Learning, Time Series Analysis, Probabilistic Modeling, Generative Modeling

Education

University of Hildesheim — Hildesheim, Germany
Ph.D. in Data Analytics — 2025
Summa cum Laude

Nanyang Technological University, Singpaore — Singpaore
Master of Engineering (by research) in Electrical and Electronic Engineering — 2017

Jawaharlal Nehru Technological University Anantapur — Anantapur, India
Bachelor of Technology in Electrical and Electronics Engineering — 2012
First Class with Distinction

Experience

Information Systems and Machine Learning Lab — University of Hildesheim, Germany
Post Doctoral Researcher — 03.2025-

- Coordinator of Data Analysis and Research Center
- Supporting PhD students in their research
- Teaching lectures on Machine Learning related topics

Pay Per Click Group — Booking.com, Netherlands
Machine Learning PhD research intern — 07.2024-12.2024

- Coordinator of Data Analysis and Research Center
- Supporting PhD students in their research
- Teaching lectures on Machine Learning related topics

Information Systems and Machine Learning Lab — University of Hildesheim, Germany
Research Assistant — 08.2018-07.2024

- Project Manager of IIP-EcoSphere
- Supervising Master students in their thesis
- Teaching Assistance for lectures on Machine Learning related topics

Department of Computer Science — Nanyang Technological University, Singapore
Research Associate — 07.2016-01.2018

- Research in data mining in social and information networks

- Supporting Bachelor's students for their projects
- Papers in IEEE BigData 2018, CIKM 2017

Central Scientific Instruments Organization Council of Scientific and Industrial Research, India
Trainee Scientist 09.2012–08.2014

- Research and Development of Energy Management Systems for commercial buildings
- Energy Auditing for office workspaces

Honors & Awards

- Sebastian Thrun Prize for best computer science student, *University of Hildesheim* 2025
- Faculty 4 best dissertation award, *University of Hildesheim* 2025

Publications

[1] V. K. Yalavarthi, R. Scholz, C. Klötergens, K. Madhusudhanan, S. Born, and L. Schmidt-Thieme, "Reliable probabilistic forecasting of irregular time series through marginalization-consistent flows," in *The Fourteenth International Conference on Learning Representations*, 2026.

[2] V. K. Yalavarthi, R. Scholz, S. Born, and L. Schmidt-Thieme, "Probabilistic Forecasting of Irregularly Sampled Time Series with Missing Values via Conditional Normalizing Flows," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[3] C. Klötergens, V. K. Yalavarthi, R. Scholz, M. Stubbemann, S. Born, and L. Schmidt-Thieme, "Physiome-ODE: A Benchmark for Irregularly Sampled Multivariate Time-Series Forecasting Based on Biological ODEs," in *The Thirteenth International Conference on Learning Representations*, 2025.

[4] N. Ahmed, V. K. Yalavarthi, and L. Schmidt-Thieme, "Motif-aware Graph Neural Networks for Networked Time Series Imputation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[5] K. Madhusudhanan, V. K. Yalavarthi, J. Sonntag, M. Stubbemann, and L. Schmidt-Thieme, "TabResFlow: A Normalizing Spline Flow Model for Probabilistic Univariate Tabular Regression," 28th European Conference on Artificial Intelligence (ECAI), 2025. arXiv: `2508.17056 [cs]`.

[6] V. K. Yalavarthi et al., "Grafiti: Graphs for forecasting irregularly sampled time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

[7] C. Klötergens, V. K. Yalavarthi, M. Stubbemann, and L. Schmidt-Thieme, "Functional Latent Dynamics for Irregularly Sampled Time Series Forecasting," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD)*, 2024. DOI: `10.1007/978-3-031-70359-1_25`.

[8] V. K. Yalavarthi, J. Burchert, and L. Schmidt-Thieme, "Tripletformer for probabilistic interpolation of irregularly sampled time series," in *2023 IEEE International Conference on Big Data (BigData)*, IEEE, 2023.

[9] S. Jawed, K. Madhusudhanan, V. K. Yalavarthi, and L. Schmidt-Thieme, "Forecasting Early with Meta Learning," in *2023 International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2023.

[10] V. K. Yalavarthi, J. Burchert, and L. Schmidt-Thieme, "DCSF: Deep convolutional set functions for classification of asynchronous time series," in *2022 IEEE 9th International Conference on Data Science and Advanced Analytics (DSAA)*, IEEE, 2022.

[11] T. Akar, T. Werner, V. K. Yalavarthi, and L. Schmidt-Thieme, "Open Set Recognition for Time Series Classification," in *Advances in Knowledge Discovery and Data Mining*, 2022.

[12] V. K. Yalavarthi, J. Grabocka, H. Mandalapu, and L. Schmidt-Thieme, "Gait verification using deep learning with a pairwise loss," in *2019 International Conference of the Biometrics Special Interest Group (BIOSIG)*, IEEE, 2019.

[13] V. K. Yalavarthi and A. Khan, "Steering top-k influencers in dynamic graphs via local updates," in *2018 IEEE International Conference on Big Data (Big Data)*, IEEE, 2018.

[14] X. Ke, M. Teo, A. Khan, and V. K. Yalavarthi, "A demonstration of perc: Probabilistic entity resolution with crowd errors," *Proceedings of the VLDB Endowment*, 2018.

[15] M. J. Er, V. K. Yalavarthi, N. Wang, and R. Venkatesan, "A Novel Incremental Class Learning Technique for Multi-class Classification," in *Advances in Neural Networks – ISNN 2016*, 2016.

Technical Skills

Programming Languages: Python, Tensorflow, PyTorch
Languages: Telugu (native), English (fluent), Hindi (working proficiency), German (basic)

Teaching

Course Instructor

- Machine Learning - 1, *University of Hildesheim* .... WiSe 2025-26
- Machine Learning - 2, *University of Hildesheim* SoS2 2025

Teaching Assistance

- Deep Learning, *University of Hildesheim* .... SoSe 2019
- Deep Learning, *University of Hildesheim* .... SoSe 2020
- Modern Optimization Techniques, *University of Hildesheim* .... WiSe 2022-23

Service

Peer Review

- ICLR: 2025,2026
- NeurIPS: 2025
- AAAI: 2026